

String

1. Given the string `"Monty Python"`:

- (a) Write an expression to print the first character.
- (b) Write an expression to print the last character.
- (c) Write an expression including `len` to print the last character.
- (d) Write an expression that prints `"Monty"`.

2. Given the string `"homebody"`:

- (a) Write an expression using slicing to print `"home"`.
- (b) Write an expression using slicing to print `"body"`.

3. Given a variable `s` containing a string of even length:

- (a) Write an expression to print out the first half of the string.
- (b) Write an expression to print out the second half of the string.

4. Given a variable `s` containing a string of odd length:

- (a) Write an expression to print the middle character.
- (b) Write an expression to print the string up to but not including the middle character (i.e., the first half of the string).
- (c) Write an expression to print the string from the middle character to the end (not including the middle character).

5. Given `x = 'water'`, what is returned by `x.replace('w','c',1)`?

6. Given the string `s = "What is your name?"`:

- (a) What is returned by `s[:2]`?
- (b) What is returned by `s[2:8:-1]`?

7. Given the string variable `x = 'acegikmoqsuwy'` and `y = '+bdfhjlnprtvxz'`, use indexing to create a string `z` that is the lowercase English alphabet.

8. The plus sign (+) is *overloaded* in Python. Explain why `5 + 4` equals 9, `'5' + '4'` equals `'54'`, and `5 + 4.0` equals 9.0.

9. What will be printed by the following?

```
x = 'This is a test.'
print(x * 3)
```

10. In the following program, replace the **for** with a **while** loop.

```
S="I had a cat named amanda when I was little"
count = 0
for i in S:
    if i == "a":
        count += 1
print(count)
```

11. (String operators) The Monty Python comedy troupe has a famous skit set in a restaurant whose menu is predominately Spam—a canned meat mixture of ham and pork. One menu entry was “Spam, Spam, Spam, Spam, Spam, baked beans, Spam, Spam, Spam, and Spam.” Write a Python string expression using both the concatenation (+) and repetition (*) string operators to form that menu entry.

12. The following Python statement generates this error: "ValueError: too many values to unpack." Why?

```
first,second = input('two space-separated numbers:')
```

13. We know that writing the following code:

```
print("I like writing in Python.")
print("It is so much fun.")
```

will result in:

```
I like writing in Python.
It is so much fun.
```

when executed. However, can you manage to do this same task with only one line of code?

14. Five string methods manipulate case: `capitalize`, `title`, `swapcase`, `upper`, and `lower`. Consider the strings: `s1 = "concord"`, `s2 = "souix city"`, `s3 = "HONOLULU"`, and `s4 = "TopHat"`.

- (a) Describe what `capitalize` does.
- (b) Describe what `swapcase` does.
- (c) Describe what `upper` does.
- (d) Describe what `lower` does.
- (e) Describe what `title` does.

15. It is possible to combine string methods in one expression. Given the expression `s="CAT"`, what is `s.upper().lower()` ?

16. Two string methods `left` and `right` justify strings within a specified width. In addition, they default by filling in with spaces but can be specified to fill in with a character. Consider `s = "Topkapi"` and `s.rjust(20, ".")` or `s.ljust(15)`. Experiment with right and left justification. Describe the rules for what `ljust` and `rjust` do.

17. Two string methods find where a character is in a string: `find` and `index`.

- (a) Both work the same if a character is found, but they behave differently if the character is not found. Describe the difference in how they behave when the character is not found.
- (b) The `find` and `index` methods are not limited to finding single characters. They can search for substrings. Given `s = "Topkapi"`, what does `s.find("kap")` print? Describe the rule for what `find` prints.

18. Using the `input` command, prompt for input and then convert the input to lowercase.

19. Convert a string that is all capitals into a string where only the first letters are capitals. For example, convert `"NEW YORK"` to `"New York"`.

20. Experiment with the `count` method. What does it count?

For example,

```
some string = "Hello world!"
some string.count("o")
```

21. Experiment with the `strip` method. What does it do?

For example,

```
some string = "Hi!....."
some string.strip("!.")
```

- 22.** The string methods that start with “is” all return either `True` or `False`. Experiment with them to figure out how they work—i.e., what causes them to return “True” and what causes them to return “False.”
- 23.** Let, `name_str = 'Albert Einstein'`. How would you extract the first name and last name from `name_str` using string operator `:`?
- 24.** In British English, there is the word *flavour*. The American spelling is “flavor”. Suppose you have a string in Python called `brit_word = 'flavour'` and you want to convert it into the American variant and store it in a string called `amer_word`. How would you do it?
- 25.** Which of the following works without any error?
- (a) `var = 'xyz' * 10.5`
 - (b) `var = 'xyz' * '5'`
 - (c) `var = 'xyz' * 5`
 - (d) `var = 'xyz' * 5.0`
- 26.** (Reversing a string) Given a string variable `X = 'Alan Turing'`, write an expression to reverse it to get string `Y = 'gniruT nalA'`.
- 27.** Suppose you have a string `ab_string = 'abababababababab'`. Write an expression to remove all the *b*’s and create a string `a_string = 'aaaaaaaa'`.
- 28.** Given the string `'abcdefghijkl'`, write a single line of code that will print the following (Hint: Slicing is your friend):
- (a) `'jihgfedcba'`
 - (b) `'adgj'`
 - (c) `'igeca'`
- 29.** Using the `find` method, write a short program that will print out the index of both *o*’s when given the input “Who’s on first?”
- 30.** Write a program that given a name in the form of “Chapman, Graham Arthur” will convert it to the form “Graham Arthur Chapman.”
- 31.** The expression `'dog' + 's'` will return `'dogs'`. What is returned by the expression `'dog' - 'g'`? Explain.

List

1. Write a program that finds the maximum value of the given list, assuming that the list contains at least one element.

Try your program with the following array

2	4	7	4	23	5	1	4	8	9
---	---	---	---	----	---	---	---	---	---

2. Write a program that calculates the average value of the given list.

Try your program with the following list

4	7	1	5	11	53	12	46	84	23
---	---	---	---	----	----	----	----	----	----

3. Write a program that prints the given list of integers in reverse order.

Try your program with the following list

2	6	7	45	23	53	14	45	89	5
---	---	---	----	----	----	----	----	----	---

4. Write a program that accepts two lists of integers and prints true if each element in the first list is less than the element at the same index in the second list. Your program should print false if the lists are not the same length.
5. Write a program that accepts a list of integers and two indexes and swaps the elements at those indexes
6. Write a program that accepts two lists of integers and prints a new list containing all elements of the first list followed by all elements of the second.
7. Write a program that accepts a list of integers and an integer value as its parameters and prints the last index at which the value occurs in the list. The program should print -1 if the value is not found. For example, in the list [74, 85, 102, 99, 101, 85, 56], the last index of the value 85 is 5.
8. Write a program that prints the range of values in a list of integers. The range is defined as 1 more than the difference between the maximum and minimum values in the list. For example, if a list contains the values [36, 12, 25, 19, 46, 31, 22], the program should return 35. You may assume that the list has at least one element.
9. Write a program that accepts a list of integers, a minimum value, and a maximum value and prints the count of how many elements from the list fall between the minimum and maximum (inclusive). For example, in the list [14, 1, 22, 17, 36, 7, -43, 5], for minimum value 4 and maximum value 17, there are four elements whose values fall between 4 and 17.
10. Write a program that accepts a list of real numbers and prints true if the list is in sorted (nondecreasing) order and false otherwise. For example, if lists named list1 and list2 store [16.1, 12.3, 22.2, 14.4] and [1.5, 4.3, 7.0, 19.5, 25.1, 46.2] respectively, the program should print false for list1 and true for list2 respectively. Assume the list has at least one element. A one-element list is sorted.